

Khalid Alduwaysan

AI Engineer | Data Scientist

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Education

King Saud University Bachelor of Computer Science • GPA: 4.37 - Second Class Honors	08/2020 – 05/2025
Udacity Nanodegree Program: Foundation of Generative AI	12/2024 – 02/2025

Experience

SDA - Saudi Digital Academy Software Engineering Bootcamp Intern • Developed full-stack web apps with Python, Django, React, and PostgreSQL. • Built interactive front-ends and integrated APIs with efficient database management	08/2025 – Present
King Salman Global Academy for Arabic Language Artificial Intelligence Intern • Developed "Raqim", a hybrid Arabic OCR correction system combining Tesseract, corpus filtering, and LLMs. • Benchmarked multiple LLMs (Gemini Flash 2.0, GPT-4, Mistral, LLaMA) and integrated the top performer. • Created synthetic training data via txt2img pipeline to enhance recognition across fonts and layouts.	01/2025 – 06/2025
SDAIA - Saudi Data & AI Authority Artificial Intelligence and Data Science T5 Bootcamp Intern • Built and optimized ML/DL models (ANN, CNN, LSTM, YOLO) for classification, object detection, and forecasting. • Applied NLP techniques (CBOW, Arabic text classification) and built a RAG system with LangChain for contextual Q&A. • Conducted EDA, clustering, and dimensionality reduction on large datasets to extract insights and improve model performance.	07/2024 – 10/2024

Courses & Certificates

Efficient Large Language Model (LLM) Customization NVIDIA Tuwaiq • Covers advanced LLM customization techniques, including PEFT, prompt engineering, LoRA, QLoRA, pruning, and RAG.	07/2025 – 07/2027
Building Transformer-Based NLP Applications NVIDIA Tuwaiq • "Skilled in Transformers, MLM/NSP, tokenization, fine-tuning, and building NLP solutions with NVIDIA."	08/2025 – 08/2027

Projects

OCR Correction Using Dictionary-Based Methods and LLMs • Leveraged Large Language Models (LLMs) and rule-based post-processing techniques to elevate Tesseract Arabic OCR output accuracy, achieving a 15.47% in CER, and raising the BLEU score to 88.32%.	01/2025 – 03/2025
Motorbike Violation Monitoring System using Computer Vision • Developed an AI-driven motorbike violation detection system using YOLO model achieving over 70% mAP50 objects detection rate, and enabling accurate OCR-based license plate extraction.	09/2024 – 10/2024
Household Price Tracker Graduation Project • Developed a website to help customers monitor and compare product prices across supermarkets by automating data collection through web scraping. Pre-processed and normalized product data, applying NLP techniques such as fuzzy matching and TF-IDF to identify equivalent items across different retailers.	08/2024 – 12/2024

Skills

• Python	• SQL	• MongoDB	• PyTorch
• Scikit-Learn	• LLM	• LangChain	• Docker
• Git	• FastAPI	• Computer Vision	• Apache Airflow

References

Rawan Al-Matham, ANLP Research Manager,
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Languages

Arabic	English
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